

1 Compact spatial realization of the changed two-stage trajectory

We now realize the actual damped two-stage trajectory, including its changed first threshold, inherited scale, second threshold, selected second return, and evolving finite hold, in the full spatial equations. The coefficients used here are those constructed in the direct return theorem and Theorem 1. The present calculation supplies their entire spatial realization; in particular it does not interpret a modal damping equation as an unforced equation for the released profile. The physical equations and orientation are

$$\begin{aligned} \theta_t + u \cdot \nabla \theta &= d\Delta\theta + f_\theta, \\ u_t + (u \cdot \nabla)u &= d\Delta u + \theta e_2 + f_u, & \operatorname{div} u &= 0, & p &= 0, \\ J &= \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, & e_2 &= (0, 1), & x &\in \mathbb{R}^2. \end{aligned}$$

Both physical coefficients equal the explicitly selected $d > 0$. The original parameters remain

$$A_0 \geq 1, \quad \mu = \lambda_0 = \lceil \sqrt{A_0}/2 \rceil, \quad \nu = \sqrt{A_0}, \quad K_j = \mu \lambda_j \quad (j = 1, 2).$$

Thus d is not the source time-scaling parameter ν . Here λ_j denotes exactly the source frequency written $\hat{\lambda}_j$ in the preceding entry calculation.

1.1 The actual coefficients and the nested material regions

Write $e(s) = (\sin s, \cos s)$, $e_0 = R_\alpha e_2$, $p_1 = e(s_*)$, and $p_2 = e(s)$, where $s = L_2\nu/\sigma$, $a = \sin s$, $h_0 = \cos s$, and $\Gamma = \sigma a$ use the actual diffusive inherited scale σ . The activation functions are

$$w_1 = \eta(\Gamma_1(t - t_{01})), \quad w_2 = \eta(\Gamma(t - t_{02})), \quad \Gamma_1 = \nu \sin s_*, \quad t_{01} = 1/\nu.$$

The source step η is smooth nondecreasing, zero on $(-\infty, 0]$ and one on $[1, \infty)$. All inactive increments below are defined to be zero. The finite coefficient construction has the exact equations

$$G_{<j} = -A_0 e_0 + \sum_{i < j} w_i K_i \Theta_i \zeta_i, \quad D_{<j} = \dot{\alpha} J + \sum_{i < j} \frac{w_i \Omega_i}{r_i^2} J \zeta_i \otimes \zeta_i, \quad (1)$$

$$\begin{aligned} \dot{M}_j &= D_{<j} M_j, & M_j(t_{0j}) &= I, & \zeta_j &= M_j^{-T} p_j, \quad r_j^2 = |\zeta_j|^2, \\ \dot{\Theta}_j &= -\frac{J \zeta_j \cdot G_{<j}}{K_j r_j^2} \Omega_j - d K_j^2 r_j^2 \Theta_j, & \dot{\Omega}_j &= K_j \zeta_{j,1} \Theta_j - d K_j^2 r_j^2 \Omega_j. \end{aligned} \quad (2)$$

The common angle is the selected original first control followed by the exact second feedback control. In particular the first threshold time is $t_{01} + L_1/(\Gamma_1 - dK_1^2)$, not the inviscid threshold time. The source first transition and pulse are translated to that actual threshold. During second growth and transition, $\alpha = s_*$, $M_2 = I$, $\Omega_1 = 0$, and Θ_1 diffuses. The second activation time is exactly $t_{02} = t_{b1}$, the actual first return time, with no intervening gap. During second steering and holding the exact inverse is

$$\begin{aligned} M_1 &= R_\alpha, \quad M_2 = R_{\alpha-s_*} \begin{pmatrix} 1 & -(h - h_0)/a \\ 0 & 1 \end{pmatrix}, \quad \Omega_1 = \sigma W, & \Theta_1 &= -\sigma^2 B/K_1, & (3) \\ h' &= W, \quad W' = B \frac{Y - hz}{h^2 + a^2} - \delta_1 W, & B' &= -C_1 W - \delta_1 B, \quad Y = \sqrt{h^2 + a^2 - a^2 z^2}, \end{aligned}$$

where $\tau = \Gamma(t - t_1)$, a prime denotes $d/d\tau$, $\delta_1 = dK_1^2/\Gamma$, $C_1 = A_0 \sin s_*/(\sigma^2 a)$, and z is the selected source ramp, pulse, and zero hold. The proved bounds give

$$h_0 \leq h < 3, \quad 0 \leq W < 9, \quad 1/4 < B \leq 2, \quad h_0 \geq \sqrt{63}/8, \quad r_1 = 1, \quad h_0 \leq r_2 < \sqrt{10}.$$

These are actual changing coefficients, not independently chosen affine backgrounds. Every $D_{<j}$ has trace zero: the rotation has trace zero and $\text{tr}(J\zeta_i \otimes \zeta_i) = \zeta_i \cdot J\zeta_i = 0$. Jacobi's determinant formula therefore gives $\det M_j = 1$. The phase equation follows by differentiating the inverse transpose: $\dot{\zeta}_j = -D_{<j}^T \zeta_j$.

On $[-\pi, \pi]$ define the released profile by the following formula and then extend it periodically to $\mathbb{R}$:

$$F(s) = \sin s + \chi(s)(s - \sin s), \quad P' = F, \quad \int_{-\pi}^{\pi} P(s) ds = 0.$$

Here χ is even, supported in $(-\pi, \pi)$, and equals one on $[-s_0, s_0]$, with $0 < s_0 < 1$. Oddness and periodicity give a unique even periodic primitive P with the specified mean. In particular $F(s) = s$ on $|s| \leq s_0$; its primitive there is $P(0) + s^2/2$, with its original constant $P(0)$ retained. Let $f \in C_c^\infty(B_1)$ be radial, $0 \leq f \leq 1$, and $f = 1$ on $B_{1/2}$. Retain the source radii

$$\ell_1 = \frac{s_0}{C_\ell \mu}, \quad \ell_2 = \frac{\lambda_1^{-3}}{\mu}, \quad C_\ell \geq 4.$$

Define the material coordinates and envelopes

$$y_j = M_j^{-1}x, \quad g_j(x, t) = f(y_j/\ell_j), \quad s_j = K_j \zeta_j \cdot x.$$

With $N = 1 + (3 - h_0)/a$, the matrix expression (3) and $\|I + qE_{12}\| \leq 1 + |q|$ show

$$\|M_1\| = \|M_1^{-1}\| = 1, \quad \|M_2\|, \|M_2^{-1}\| \leq N. \quad (4)$$

The second inequality also holds before steering, where $M_2 = I$. The entry construction, (83), selects λ_1 with the explicit strict inequalities

$$N\lambda_1^{-3} < \frac{s_0}{2C_\ell}, \quad N\lambda_1^{-2} < s_0. \quad (5)$$

Consequently every point of the second support satisfies $|x| \leq N\ell_2 < \ell_1/2$ and $|K_1 \zeta_1 \cdot x| \leq K_1 N \ell_2 < s_0$. It therefore lies in the exact affine cell of the first wave for the whole construction. The first support lies in B_{ℓ_1} ; since $\mu \ell_1 = s_0/C_\ell < s_0$, the base is exactly affine there for every angle. Strictness leaves an open neighborhood of each support on which the corresponding affine identities hold.

1.2 Original datum, fields, and complete residual forces

Retain the original radial cutoff $\chi_0 \in C_c^\infty(B_{R_0})$, equal to one on B_2 , and the exact datum

$$\theta_{\text{in}}(x) = -\frac{A_0}{\mu} \sin(\mu x_2) \chi_0(x), \quad u_{\text{in}} = 0.$$

Choose the radial $H_0 \in C_c^\infty(B_{R_H})$ used in the base construction, where $R_H > R_0 + 1$ and $H_0 = |x|^2/2$ on B_{R_0+1} . Put

$$\begin{aligned} \theta_0 &= -\frac{A_0}{\mu} F(\mu e_0 \cdot x) \chi_0(x), & \theta_{\text{ref}} &= -\frac{A_0}{\mu} F(\mu x_2) \chi_0(x), \\ \theta_b &= \theta_0 + (1 - \eta(\nu t))(\theta_{\text{in}} - \theta_{\text{ref}}), & \psi_b &= \dot{\alpha} H_0, & u_b &= J \nabla \psi_b. \end{aligned} \quad (6)$$

The angle is zero before $1/\nu$; the initial interpolation is therefore exact and the initial velocity is zero. Define

$$\begin{aligned} T_j &= w_j \Theta_j, & Q_j &= \frac{w_j \Omega_j}{K_j^2 r_j^2}, & V_j &= T_j F(s_j) g_j, & \Psi_j &= Q_j P(s_j) g_j, \\ v_j &= J \nabla \Psi_j, & \theta &= \theta_b + V_1 + V_2, & u &= u_b + v_1 + v_2. \end{aligned} \quad (7)$$

The flat activation functions make the zero extensions of the increments smooth, with all spatial and temporal derivatives, at $t = t_{0j}$. The velocity is divergence free because mixed partial derivatives commute.

Hereafter fix a layer and temporarily suppress its index. Write $k = K\zeta$, $\rho = |k|^2 = K^2 r^2$, $D = D_{<j}$, $G = G_{<j}$, and $b = k_1$. All functions of one variable in the formulas below are evaluated at $s = k \cdot x$. Define

$$Q_* = \frac{\dot{w}\Omega + wb\Theta}{\rho} - Q\frac{\dot{\rho}}{\rho}, \quad \dot{\rho} = -2k \cdot D^T k. \quad (8)$$

Thus $\dot{Q} = Q_* - d\rho Q$, directly from (2), without a constant-phase-length assumption. The complete added scalar and vector forces are

$$E_\theta = \dot{w}\Theta Fg + QP(J\nabla g \cdot G) + QT(F^2 - PF')g(Jk \cdot \nabla g) - dT\{\rho(F + F'')g + 2F'k \cdot \nabla g + F\Delta g\}, \quad (9)$$

$$E_u = 2Dv + Q_*J\{kFg + P\nabla g\} + (\nabla v)v - TFge_2 - d(\rho v + \Delta v). \quad (10)$$

These are full vector forces at the declared pressure $p = 0$. For completely componentwise evaluation one uses

$$v = QJ\{kFg + P\nabla g\}, \quad \nabla v = QJ\{k \otimes kF'g + F(k \otimes \nabla g + \nabla g \otimes k) + P\nabla^2 g\}, \quad (11)$$

$$\Delta V = T\{\rho F''g + 2F'k \cdot \nabla g + F\Delta g\}, \quad (12)$$

$$\Delta v = QJ\{\rho kF''g + \rho F'\nabla g + 2F'k(k \cdot \nabla g) + 2F(\nabla^2 g)k + Fk\Delta g + P\nabla\Delta g\}. \quad (13)$$

Thus every phase, envelope, and mixed Laplacian term is displayed. The base forces are

$$f_{\theta,b} = \nu\eta'(\nu t)(\theta_{\text{ref}} - \theta_{\text{in}}) - d\Delta\theta_b, \quad f_{u,b} = \ddot{\alpha}J\nabla H_0 + \dot{\alpha}^2(\nabla(J\nabla H_0))J\nabla H_0 - \theta_b e_2 - d\dot{\alpha}J\nabla\Delta H_0. \quad (14)$$

The requested total forces are exactly

$$f_\theta = f_{\theta,b} + E_{\theta,1} + E_{\theta,2}, \quad f_u = f_{u,b} + E_{u,1} + E_{u,2}. \quad (15)$$

Proof of the full spatial equations. For the material derivative $\mathcal{D} = \partial_t + Dx \cdot \nabla$, differentiating $M^{-1}x$ gives $\mathcal{D}(M^{-1}x) = 0$. The phase equation gives $\mathcal{D}(k \cdot x) = 0$ as well. Consequently $\mathcal{D}g = 0$, $\mathcal{D}V = \dot{T}Fg$, and $\mathcal{D}\Psi = \dot{Q}Pg$. On this support the actual older temperature and velocity are $G \cdot x$ and Dx by (5); in particular their gradient and Jacobian are exactly G, D , even while these change. The temperature advection increment is therefore $\dot{T}Fg + v \cdot G + v \cdot \nabla V$. Using $Jk \cdot k = 0$ and $J\nabla g \cdot \nabla g = 0$ gives

$$v \cdot \nabla V = QT(F^2 - PF')g(Jk \cdot \nabla g), \quad v \cdot G = QFg(Jk \cdot G) + QP(J\nabla g \cdot G).$$

Since $\dot{\Theta} = -(Jk \cdot G)\Omega/\rho - d\rho\Theta$, the first term in $v \cdot G$ cancels the coupled part of $\dot{T}Fg$ exactly. Subtraction of $d\Delta V$, computed by the product rule, proves (9).

For every trace-free two by two matrix, direct multiplication gives $DJ + JD^T = 0$. Differentiation of a gradient under $\mathcal{D}$ therefore gives

$$\mathcal{D}(J\nabla\Psi) = DJ\nabla\Psi + J\nabla(\mathcal{D}\Psi).$$

The other linear advection increment is $(v \cdot \nabla)(Dx) = Dv$. Adding it to this formula, adding the self-advection $(\nabla v)v$, and subtracting $Ve_2 + d\Delta v$ proves (10), after using $\dot{Q} = Q_* - d\rho Q$. Differentiating $\Delta\Psi = Q(\rho F'g + 2Fk \cdot \nabla g + P\Delta g)$ and applying J proves all six groups in (13).

The base phase $\mu e_0 \cdot x$ is transported by $\dot{\alpha}Jx$, and its radial cutoff is also transported. Before $1/\nu$ its velocity is zero. This proves the scalar base formula. Differentiating $\dot{\alpha}J\nabla H_0$ proves the vector base formula directly on all of $\mathbb{R}^2$. Finally, adding increments in the order 1, 2 accounts

exactly for both directions of each interaction. In the second increment, $\mathcal{D}$ contains $u_b + v_1$, and the terms $v_2 \cdot G_{<2}$ and $D_{<2}v_2$ contain the reverse interactions with that full older state. Outside an increment's support all its derivatives vanish. Thus the residual identities telescope globally to (15), with no omitted interaction and no inverse curl or pressure reconstruction. $\square$

The material-coordinate form makes the original metric equally explicit. Set $C = M^{-1}M^{-T}$ and $\tilde{g}(y) = f(y/\ell)$. The pullback of the spatial derivative and Laplacian on a function of (s, y) are

$$\mathcal{P} = K\zeta\partial_s + M^{-T}\nabla_y, \quad \mathcal{L} = K^2r^2\partial_s^2 + 2K(Cp \cdot \nabla_y)\partial_s + \sum_{l,m} C_{lm}\partial_{y_l}\partial_{y_m}. \quad (16)$$

Indeed $s = Kp \cdot y$, $\nabla_x = M^{-T}\nabla_y$ on a material function, and $|M^{-T}p|^2 = p^T Cp = r^2$; multiplying the two derivative operators proves (16). For example $k \cdot \nabla_x g = K(Cp) \cdot \nabla_y \tilde{g}$. This proves an exact coordinate correspondence for every displayed mixed derivative and keeps the source radii in their original units.

1.3 What the affine cells and sine regions actually satisfy

On $g = 1$ and $|s| < s_0$ the wave is exactly

$$V = Tk \cdot x, \quad v = QJk \otimes kx = \frac{w\Omega}{r^2}J\zeta \otimes \zeta x.$$

Both Laplacians vanish, but their damping contributions to the forces (9)–(10) are

$$-d\rho V, \quad -d\rho v. \quad (17)$$

They do not vanish unless the corresponding wave vanishes. For the first wave the scalar core force is $-dK_1^2(w_1K_1\Theta_1\zeta_1) \cdot x$ after activation, and the curl of its damping vector force is $-dK_1^2w_1\Omega_1$. Thus the evolving affine parent itself has the force required by its changed amplitude equations. On an uncut region where $F(s) = \sin s$, instead, $F'' = -F$ and the two damping contributions are zero: $\rho V + \Delta V = 0$ and $\rho v + \Delta v = 0$. The non-damping vector terms in (10) are still present; this cancellation is not a claim of unforced momentum at $p = 0$. In cutoff regions every term in (12)–(13) must also be kept. These restrictions prove exactly how the damped coefficient dynamics map into the released spatial profile. They differ from retaining an inviscid state and adding only $-d\Delta$ of that state: the present amplitudes, parent, phases and controls have all changed according to (2).

1.4 Explicit bounds at every spatial derivative order

For a scalar, vector, or tensor field U , let $|U|_m = \sup_x \|D_x^m U(x)\|$, using Euclidean multilinear operator norms and $m \geq 0$. For a one-variable profile W write $[W]_m = \sup_s |W^{(m)}(s)|$, and put $F_m^f = \sup_y \|D^m f(y)\|$. Set $N_1 = 1$, $N_2 = N$, $r_{1,-} = r_{1,+} = 1$, $r_{2,-} = h_0$, $r_{2,+} = \sqrt{10}$. Define finite explicit numbers

$$\begin{aligned} \kappa_j^* &= K_j r_{j,+}, & e_{j,m} &= (N_j/\ell_j)^m F_m^f, \\ \mathcal{B}_{j,m}(W) &= \sum_{q=0}^m \binom{m}{q} (\kappa_j^*)^q [W]_q e_{j,m-q}, & \mathcal{C}_{j,m}(P) &= \sum_{q=0}^m \binom{m}{q} (\kappa_j^*)^q [P]_q e_{j,m-q+1}. \end{aligned} \quad (18)$$

Here κ_j^* is merely a frequency bound, not a diffusivity. Repeated differentiation and the chain rule give $|g_j|_m \leq e_{j,m}$, $|W(s_j)g_j|_m \leq \mathcal{B}_{j,m}(W)$, and $|P(s_j)\nabla g_j|_m \leq \mathcal{C}_{j,m}(P)$. To prove the middle inequality, assign each of the m derivatives either to the profile or to the envelope. There are $\binom{m}{q}$ assignments with q profile derivatives, each bounded by the corresponding summand. The other inequalities follow by the same argument. This proof applies to all directions in the defined operator norm, not only coordinate derivatives.

The actual threshold amplitudes are $P_j^* = (\nu^2/\mu)\lambda_j^{-7/8}$. The finite coefficient calculation, (92), supplies the following global bounds through the hold:

$$\begin{aligned} U_{\Theta,1} &= \max\{e^{2+\Lambda_1^{-1}} P_1^*, 2\sigma^2/K_1\}, & U_{\Omega,1} &= \max\{(K_1/\nu)e^{2+\Lambda_1^{-1}} P_1^*, 9\sigma\}, \\ U_{\Theta,2} &= e^{18} P_2^*, & U_{\Omega,2} &= 3K_2 U_{\Theta,2}/\sigma. \end{aligned} \quad (19)$$

In particular these preserve the actual source frequencies and the changed inherited scale. Put

$$A_1 = \sup_{[0,T_f]} |\dot{\alpha}|, \quad A_2 = \sup_{[0,T_f]} |\ddot{\alpha}|, \quad D_1^* = A_1, \quad D_2^* = A_1 + U_{\Omega,1}, \quad G_1^* = A_0, \quad G_2^* = A_0 + K_1 U_{\Theta,1}.$$

Explicit profile bounds for A_1, A_2 are given below. Set $W_{1,1} = \Gamma_1[\eta']_0$ and $W_{2,1} = \Gamma[\eta']_0$. Then

$$\begin{aligned} Q_j^* &= \frac{U_{\Omega,j}}{K_j^2 r_{j,-}^2}, \\ Q_{*,j}^* &= \frac{W_{j,1} U_{\Omega,j} + \kappa_j^* U_{\Theta,j} + 2D_j^* U_{\Omega,j}}{K_j^2 r_{j,-}^2}. \end{aligned} \quad (20)$$

The bounds $|Q_j| \leq Q_j^*$ and $|Q_{*,j}| \leq Q_{*,j}^*$ follow from (8) and $|\dot{\rho}/\rho| \leq 2\|D\|$. Define

$$\begin{aligned} \mathcal{D}_{\theta,j,m} &= dU_{\Theta,j}\{(\kappa_j^*)^2 \mathcal{B}_{j,m}(F) + 2\mathcal{B}_{j,m+2}(F)\}, \\ \mathcal{D}_{u,j,m} &= dQ_j^*\{(\kappa_j^*)^2 \mathcal{B}_{j,m+1}(P) + 2\mathcal{B}_{j,m+3}(P)\}. \end{aligned} \quad (21)$$

These bound the complete damping-plus-diffusion terms in (9)–(10). In particular they include the core damping and every envelope derivative. The two in front of $\mathcal{B}_{m+2}$ or $\mathcal{B}_{m+3}$ is the sum of the two coordinate Laplacian terms; J has operator norm one. The whole added forces satisfy the explicit inequalities

$$\begin{aligned} |E_{\theta,j}|_m &\leq W_{j,1} U_{\Theta,j} \mathcal{B}_{j,m}(F) + Q_j^* G_j^* \mathcal{C}_{j,m}(P) + \mathcal{D}_{\theta,j,m} \\ &\quad + Q_j^* U_{\Theta,j} \sum_{q=0}^m \binom{m}{q} \mathcal{B}_{j,q+1}(P) \mathcal{B}_{j,m-q+1}(F), \end{aligned} \quad (22)$$

$$\begin{aligned} |E_{u,j}|_m &\leq (2D_j^* Q_j^* + Q_{*,j}^*) \mathcal{B}_{j,m+1}(P) + U_{\Theta,j} \mathcal{B}_{j,m}(F) + \mathcal{D}_{u,j,m} \\ &\quad + (Q_j^*)^2 \sum_{q=0}^m \binom{m}{q} \mathcal{B}_{j,q+1}(P) \mathcal{B}_{j,m-q+2}(P). \end{aligned} \quad (23)$$

For the scalar nonlinear term it is enough to use its exact alternative expression $v_j \cdot \nabla V_j$; differentiation of this product proves the displayed sum. For the vector nonlinear term differentiate $(\nabla v_j) v_j$ and interchange the summation index to obtain the final sum. These arguments prove all four bounds directly from the exact formulas.

The base has equally explicit bounds. Put $c_m = |\chi_0|_m$, $h_m = |H_0|_m$, and

$$b_m(W) = \sum_{q=0}^m \binom{m}{q} \mu^q [W]_q c_{m-q}, \quad B_m^0 = \frac{A_0}{\mu} \max\{b_m(F), b_m(\sin)\}.$$

The interpolation is a convex combination when $\alpha = 0$; afterwards $|\mu e_0| = \mu$. Hence $|\theta_b|_m \leq B_m^0$. The product rule and (14) give

$$\begin{aligned} |f_{\theta,b}|_m &\leq \frac{\nu A_0}{\mu} [\eta']_0 \{b_m(F) + b_m(\sin)\} + 2dB_{m+2}^0, \\ |f_{u,b}|_m &\leq A_2 h_{m+1} + B_m^0 + 2dA_1 h_{m+3} + A_1^2 \sum_{q=0}^m \binom{m}{q} h_{q+2} h_{m-q+1}. \end{aligned} \quad (24)$$

All total force norms are bounded by adding their base and two increment bounds. Integrating any bound over its interval multiplies its constant by that interval's physical length. For example

$$\int_0^{T_f} |\text{wave damping-plus-diffusion force}_j|_m dt \leq (T_f - t_{0j}) \mathcal{D}_{\bullet,j,m}. \quad (25)$$

Here $T_f = t_1 + (1 + L_2^{-1} + 1/32)/\Gamma$ is the actual end of the finite hold and the dot denotes the corresponding scalar or vector bound. Every frequency, envelope radius and diffusion coefficient is explicit. No estimate in this section asserts a bound uniform over a sequence $K_j \rightarrow \infty$.

1.5 Control derivatives and every mixed derivative order

The constants A_1, A_2 in the spatial force costs can be bounded without differentiating an unknown shooting root. The root is a fixed number in $[0, 3]$, and differentiation below is in time. For a compression L put

$$Z_0(L) = \max\{1, 3L[\eta_2]_0\}, \quad Z_b(L) = \max\{[\eta^{(b)}]_0, 3L^{b+1}[\eta_2^{(b)}]_0\} \quad (b \geq 1).$$

These bound the derivatives of the source ramp and pulse z ; the hold has zero derivatives. During the first control, $\alpha = s_* - \arcsin(\sin s_* z)$ and $|\sin s_* z| \leq 1/4$. Twice differentiating this displayed formula gives

$$\begin{aligned} A_{1,\text{first}} &= \frac{4}{\sqrt{15}} \Gamma_1 \sin s_* Z_1(\Lambda_1), \\ A_{2,\text{first}} &= \Gamma_1^2 \left\{ \frac{4}{\sqrt{15}} \sin s_* Z_2(\Lambda_1) + \frac{64}{15\sqrt{15}} \sin^3 s_* Z_0(\Lambda_1) Z_1(\Lambda_1)^2 \right\}. \end{aligned}$$

During second steering and holding write $z_b = Z_b(L_2)$ and $y_- = \sqrt{59}/8$. The exact control is

$$\alpha_\tau = \frac{W\zeta_{1,1} - az'}{Y}, \quad Y' = \frac{hW - a^2zz'}{Y}, \quad \zeta_{1,\tau} = \alpha_\tau J\zeta_1.$$

Define the explicit constants

$$\begin{aligned} U_\alpha &= (9 + az_1)/y_-, \quad U_Y = (27 + az_1/4)/y_-, \\ U_{W'} &= \frac{128}{63}(\sqrt{10} + 3z_0) + \frac{9}{16}, \\ U_{\alpha'} &= (U_{W'} + 9U_\alpha + az_2)/y_- + (9 + az_1)U_Y/y_-^2. \end{aligned}$$

The bounds $Y \geq y_-$, $|az| \leq 1/4$, $h < 3$, $W < 9$, $B \leq 2$, $h^2 + a^2 \geq 63/64$, and $\delta_1 \leq 1/16$ give $|\alpha_\tau| \leq U_\alpha$, $|Y'| \leq U_Y$ and $|W'| \leq U_{W'}$. Differentiating the quotient for α_τ gives $|\alpha_{\tau\tau}| \leq U_{\alpha'}$. Thus one can use

$$A_1 \leq \max\{A_{1,\text{first}}, \Gamma U_\alpha\}, \quad A_2 \leq \max\{A_{2,\text{first}}, \Gamma^2 U_{\alpha'}\}.$$

On intervening growth and transition intervals the angle is constant.

For completeness the following exact finite recursion evaluates any mixed derivative, rather than treating spatial estimates as time estimates. For a fixed layer put $A = M^{-1}/\ell$ and introduce independent variables t, x, s, y . Define operators on smooth expressions in these variables by

$$\mathfrak{T} = \partial_t + (\dot{k} \cdot x) \partial_s + (\dot{A}x) \cdot \nabla_y, \quad \mathfrak{X}_i = \partial_{x_i} + k_i \partial_s + \sum_l A_{li} \partial_{y_l}. \quad (26)$$

Here ∂_t differentiates every time-dependent coefficient and ∂_{x_i} differentiates the explicitly occurring x . If $\iota^* R = R(t, x, k(t) \cdot x, A(t)x)$, the chain rule proves

$$\partial_t^b \partial_x^\gamma (\iota^* R) = \iota^* (\mathfrak{T}^b \mathfrak{X}_1^{\gamma_1} \mathfrak{X}_2^{\gamma_2} R). \quad (27)$$

Induction proves this identity for all nonnegative b, γ_i . For example, for $R = Q(t)P(s)f(y)$ it gives every streamfunction jet; applying one more $\mathfrak{X}$ and J gives every velocity jet. The commutators vanish: in $[\mathfrak{T}, \mathfrak{X}_i]$ the derivative $\dot{k}_i$ is canceled by differentiating $\dot{k} \cdot x$, and $\dot{A}_{li}$ by differentiating $(\dot{A}x)_l$. The $\mathfrak{X}_i$ commute by their displayed coefficients. This checks the order consistency of the recursion.

Apply (27) to each summand of (9)–(14), using a separate pair (s, y) for each layer when necessary. After expansion the answer is a finite sum of terms of the form

$$c(t)x^\beta \prod_{l=1}^q W_l^{(n_l)}(s_j) \prod_{l=1}^r \partial^{\gamma_l} f(y_j),$$

or the analogous base terms with χ_0, H_0 . An explicit bound for each term is obtained by replacing it by

$$\sup_{[0, T_f]} |c(t)| (N_j \ell_j)^{|\beta|} \prod_{l=1}^q [W_l]_{n_l} \prod_{l=1}^r F_{|\gamma_l|}^f,$$

and summing; for a base term use $R_H^{|\beta|}$. Each coefficient jet in c is computed by differentiating (2), the phase equations, (3), and the explicit controls. This is a finite recursion using only addition, multiplication, the stated profile derivatives and the nonzero denominators a, K_j, Γ, r_j^2, Y . Their positive lower bounds have already been proved. Induction on the requested order, applying the product and quotient rules to those equations, therefore supplies finite bounds for every coefficient jet from the preceding-order bounds. This specifies every mixed force and diffusion norm by an exact finite procedure, including the time variation of the material cutoff. It does not replace the cutoff by a stationary radial function for layer two.

1.6 Smooth finite slab, compact forces, and exact returned state

All fields and forces are smooth and supported in B_{R_H} on $0 \leq t \leq T_f$. Each velocity is the rotated gradient of a compact streamfunction, so its energy is finite. It also has the usual exact Biot–Savart relation. Indeed for $N_0(x) = (2\pi)^{-1} \log |x|$ and compact smooth $\psi = \psi_b + \Psi_1 + \Psi_2$, distributional integration by parts gives $N_0 * \Delta \psi = (\Delta N_0) * \psi = \psi$; applying $J\nabla$ gives $u = J\nabla N_0 * \omega$ with $\omega = \Delta \psi$. All integrations are legitimate because ψ is compact and N_0 is locally integrable. Odd F , even P , and even cutoffs give odd θ, u, f_θ, f_u and even ω . The pressure $p = 0$ has $p(0, t) = 0$.

At the selected second return and throughout its physical hold of duration $1/(32\Gamma)$, the coefficient calculation proves $\zeta_{2,1} = \Omega_2 = 0, w_2 = 1$, and

$$\Theta_2(t) = -P_2(t_b) \exp \left[-dK_2^2 \int_{t_b}^t |\zeta_2(r)|^2 dr \right].$$

Thus $v_2 = 0$ everywhere on this hold, while its temperature and the first parent continue changing. In the inner second affine cell the total fields have exactly

$$\theta = (-A_0 e_0 + K_1 \Theta_1 \zeta_1 + K_2 \Theta_2 \zeta_2) \cdot x, \quad u = (\dot{\alpha} J + \Omega_1 J \zeta_1 \otimes \zeta_1) x.$$

The second holding shear is zero, but the first holding shear is positive at return and evolves by the displayed system. In particular these exact full spatial fields have the origin vorticity calculated in (110):

$$\text{curl } u(0, t) = 2\dot{\alpha} + \Omega_1 = \Omega_1 \left(1 - \frac{2a^2}{h^2 + a^2} \right) \geq \frac{31}{32} \Omega_1 > 0 \quad (t_b \leq t \leq T_f). \quad (28)$$

Indeed on holding $\alpha = s_* + \arctan(a/h)$ and $\dot{h} = a\Omega_1$, so $\dot{\alpha} = -a^2\Omega_1/(h^2 + a^2)$. Moreover $h \geq h_0$, $h_0^2 + a^2 = 1$ and $a \leq 1/8$ prove the numerical lower bound. Positivity of Ω_1 at return persists because the holding equation has positive forcing B/r_2 . These are the actual final states realized by the full spatial equations.

There is also an honest compact extension of the forces in time. For $t < 0$ extend the initial state by its constant values, with $\alpha = 0$ and both increments zero. Flatness of η makes the formulas smooth at zero. To the right, continue $z = 0$ and the exact holding ODE for a collar of physical length $\epsilon_R = 1/(128\Gamma)$. Its extension is quantitative: the total scaled duration is at most $17/16$ since $L_2 \geq 64$; the same integrating-factor estimates give

$$0 \leq W \leq 26/3, \quad h \leq 1 + 16/15 + (26/3)(7/128) = 813/320 < 3, \quad B > 191/512 > 1/4.$$

For the last bound use $B_{\text{in}} \geq 1/2$, $\delta_1 \leq 1/16$, $C_1 \leq 1/32$ and $B = e^{-\delta_1 \tau} B_{\text{in}} - C_1 \int_0^\tau e^{-\delta_1(\tau-r)} W(r) dr$. These strict bounds close the positivity bootstrap and keep all denominators away from zero; the smooth finite-dimensional vector field therefore continues through this collar. The support bounds continue as well. Extend the newest temperature by its exact hold exponential, and the angle and matrices by their exact inverse. In particular no nonstationary endpoint is frozen.

Choose $\epsilon_L = 1/\nu$ and multiply the extended residual forces, not the fields on their required slab, by

$$\beta(t) = \eta(1 + t/\epsilon_L)\eta(1 + (T_f - t)/\epsilon_R).$$

This function is one on $[0, T_f]$, zero outside $(-\epsilon_L, T_f + \epsilon_R)$, and flat at the two external endpoints. The resulting forces belong to $C_c^\infty(\mathbb{R}^2 \times \mathbb{R})$ and still give exactly the solution already proved on $[0, T_f]$. On the collar the same mixed-derivative recursion applies, and the product rule includes the explicit additional factors $\epsilon_L^{-b}[\eta^{(b)}]_0$ and $\epsilon_R^{-b}[\eta^{(b)}]_0$. No equation for the continued fields outside the required slab is claimed after this force cutoff. This completes the compact finite spatial realization and its full derivative and support costs.